

Formulating the World as an MDP

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This document presents a formalization of a daily student scenario modeled as a **Markov Decision Process (MDP)**. Following the foundational theory, we perform a manual execution of the **Value Iteration** algorithm for exactly two iterations to compute optimal state values.

1. Scenario Description & MDP Formulation

We model a student's daily lifecycle on a campus. At any given decision epoch, the student evaluates their cognitive and physical state to decide whether to invest energy into academic tasks, rest to recover capacity, or engage in social leisure. The goal is to maximize long-term academic success while preventing burnout.

1.1 State Space (S)

The state space consists of three distinct, mutually exclusive operational states:

- s_1 = **Exhausted**: Low energetic capacity, high cognitive fatigue.
- s_2 = **Energetic**: Fully recovered, optimal focus, prime condition for learning.
- s_3 = **Socially Starved**: High isolation or academic burnout, demanding social connection.

1.2 Action Space (A)

From any state, the student can choose from three foundational actions:

- a_1 = **Study**: Dedicate hours to intense focus, assignments, or lectures.
- a_2 = **Sleep**: Disengage and sleep to reset mental and physical baselines.
- a_3 = **Hang Out**: Socialize with peers to satisfy psychological needs.

1.3 Transition Probabilities ($P(s' | s, a)$)

The environmental dynamics are stochastic, mapping the probability of transitioning to state s' given current state s and action a . The detailed transition matrix is structured below:

Current State (s)	Action (a)	Next State: Exhausted (s_1)	Next State: Energetic (s_2)	Next State: Socially Starved (s_3)
s_1 (Exhausted)	a_1 (Study)	1.0	0.0	0.0
	a_2 (Sleep)	0.1	0.9	0.0
	a_3 (Hang Out)	0.2	0.0	0.8
s_2 (Energetic)	a_1 (Study)	0.7	0.1	0.2
	a_2 (Sleep)	0.0	1.0	0.0
	a_3 (Hang Out)	0.1	0.3	0.6
s_3 (Socially Starved)	a_1 (Study)	0.6	0.0	0.4
	a_2 (Sleep)	0.2	0.5	0.3
	a_3 (Hang Out)	0.0	0.9	0.1

1.4 Reward Function ($R(s, a)$)

Rewards quantify immediate utility. Studying while *Energetic* yields high returns, whereas studying while *Exhausted* inflicts a penalty. Socializing and sleeping provide contextual benefits:

- $R(s_1, a_1) = -5$ (Studying while exhausted yields high frustration and minimal comprehension)
- $R(s_1, a_2) = +4$ (Sleeping when exhausted provides necessary physical relief)
- $R(s_1, a_3) = -1$ (Hanging out when exhausted causes mild physical overexertion)
- $R(s_2, a_1) = +10$ (Studying while energetic delivers optimal focus and premium academic reward)
- $R(s_2, a_2) = +2$ (Sleeping while already energetic yields minor utility)
- $R(s_2, a_3) = +4$ (Hanging out while energetic is socially fulfilling)
- $R(s_3, a_1) = -2$ (Studying while socially starved worsens academic burnout)
- $R(s_3, a_2) = 0$ (Sleeping does not solve emotional isolation)
- $R(s_3, a_3) = +8$ (Hanging out while starved provides high psychological recovery)

2. Manual Value Iteration

Value Iteration computes the optimal state values using the Bellman Optimality Equation. The update rule for step $k+1$ is:

$$V_{k+1}(s) = \max_{a \in A} [R(s, a) + \gamma \sum_{s' \in S} P(s' | s, a) V_k(s')]$$

We evaluate exactly 2 steps ($k = 1, 2$) assuming a discount factor $\gamma = 0.9$. We initialize all states to zero: $V_0(s_1) = 0$, $V_0(s_2) = 0$, $V_0(s_3) = 0$.

Iteration 1 (k = 1)

Since $V_0(s') = 0$ for all s' , the future discounted return term drops out, reducing the equation to: $V_1(s) = \max_a [R(s, a)]$.

For State s_1 (Exhausted):

- For a_1 : $-5 + 0 = -5$
- For a_2 : $4 + 0 = 4$
- For a_3 : $-1 + 0 = -1$

$$V_1(s_1) = \max(-5, 4, -1) = 4.0 \text{ (Optimal Action: } a_2 \text{)}$$

For State s_2 (Energetic):

- For a_1 : $10 + 0 = 10$
- For a_2 : $2 + 0 = 2$
- For a_3 : $4 + 0 = 4$

$$V_1(s_2) = \max(10, 2, 4) = 10.0 \text{ (Optimal Action: } a_1 \text{)}$$

For State s_3 (Socially Starved):

- For a_1 : $-2 + 0 = -2$
- For a_2 : $0 + 0 = 0$
- For a_3 : $8 + 0 = 8$

$$V_1(s_3) = \max(-2, 0, 8) = 8.0 \text{ (Optimal Action: } a_3 \text{)}$$

End of Iteration 1 Vector: $V_1 = [4.0, 10.0, 8.0]^T$

Iteration 2 (k = 2)

We use the values from V_1 to compute V_2 using $\gamma = 0.9$. Let the total value for an action be $Q_2(s, a) = R(s, a) + 0.9 \sum s' P(s' | s, a) V_1(s')$.

For State s_1 (Exhausted):

- $Q_2(s_1, a_1) = -5 + 0.9 \times [1.0(4) + 0 + 0] = -5 + 3.6 = -1.4$
 - $Q_2(s_1, a_2) = 4 + 0.9 \times [0.1(4) + 0.9(10) + 0] = 4 + 0.9 \times [0.4 + 9.0] = 4 + 0.9 \times 9.4 = 4 + 8.46 = 12.46$
 - $Q_2(s_1, a_3) = -1 + 0.9 \times [0.2(4) + 0 + 0.8(8)] = -1 + 0.9 \times [0.8 + 6.4] = -1 + 0.9 \times 7.2 = -1 + 6.48 = 5.48$
- $V_2(s_1) = \max(-1.4, 12.46, 5.48) = 12.46$ (Optimal Action: a_2)

For State s_2 (Energetic):

- $Q_2(s_2, a_1) = 10 + 0.9 \times [0.7(4) + 0.1(10) + 0.2(8)] = 10 + 0.9 \times [2.8 + 1.0 + 1.6] = 10 + 0.9 \times 5.4 = 10 + 4.86 = 14.86$
 - $Q_2(s_2, a_2) = 2 + 0.9 \times [0 + 1.0(10) + 0] = 2 + 0.9 \times 10 = 2 + 9.0 = 11.0$
 - $Q_2(s_2, a_3) = 4 + 0.9 \times [0.1(4) + 0.3(10) + 0.6(8)] = 4 + 0.9 \times [0.4 + 3.0 + 4.8] = 4 + 0.9 \times 8.2 = 4 + 7.38 = 11.38$
- $V_2(s_2) = \max(14.86, 11.0, 11.38) = 14.86$ (Optimal Action: a_1)

For State s_3 (Socially Starved):

- $Q_2(s_3, a_1) = -2 + 0.9 \times [0.6(4) + 0 + 0.4(8)] = -2 + 0.9 \times [2.4 + 3.2] = -2 + 0.9 \times 5.6 = -2 + 5.04 = 3.04$
 - $Q_2(s_3, a_2) = 0 + 0.9 \times [0.2(4) + 0.5(10) + 0.3(8)] = 0 + 0.9 \times [0.8 + 5.0 + 2.4] = 0 + 0.9 \times 8.2 = 7.38$
 - $Q_2(s_3, a_3) = 8 + 0.9 \times [0 + 0.9(10) + 0.1(8)] = 8 + 0.9 \times [9.0 + 0.8] = 8 + 0.9 \times 9.8 = 8 + 8.82 = 16.82$
- $V_2(s_3) = \max(3.04, 7.38, 16.82) = 16.82$ (Optimal Action: a_3)

End of Iteration 2 Vector: $V_2 = [12.46, 14.86, 16.82]^T$

3. Final Results Summary

The progression of state values over two complete updates of the value iteration scheme is summarized in the table below:

State (·)	Initial Value $V_0(s)$	Iteration 1 Value $V_1(s)$ (Best Action)	Iteration 2 Value $V_2(s)$ (Best Action)
s_1 (Exhausted)	0.00	4.00 (a_2)	12.46 (a_2)
s_2 (Energetic)	0.00	10.00 (a_1)	14.86 (a_1)
s_3 (Socially Starved)	0.00	8.00 (a_3)	16.82 (a_3)

Conclusion: After exactly two steps of manual optimization, the optimal policy remains stable across the horizons evaluated: sleep (a_2) when exhausted, study (a_1) when energetic, and hang out (a_3) when socially starved. The state values have absorbed rewards out to a two-step look-ahead horizon, heavily reinforcing long-term self-care and structured work cycles.